WFI pipeline

Data reduction tutorial

PHANGS-Ha Survey in a nutshell

PHANGS-Ha

PHANGS-Ha narrow-band survey is meant to complement ALMA CO maps at a comparable (cloud-scale) resolution.

Overall, we observed with seeing between 0.6” and 1.3” with a limit of 1.5” for the data to be valid.

Wide Field Imager (WFI) at MPG2.2-m La Silla

8 CCDs - FOV of 34’x33’ - 0.238 arcsec/pixel

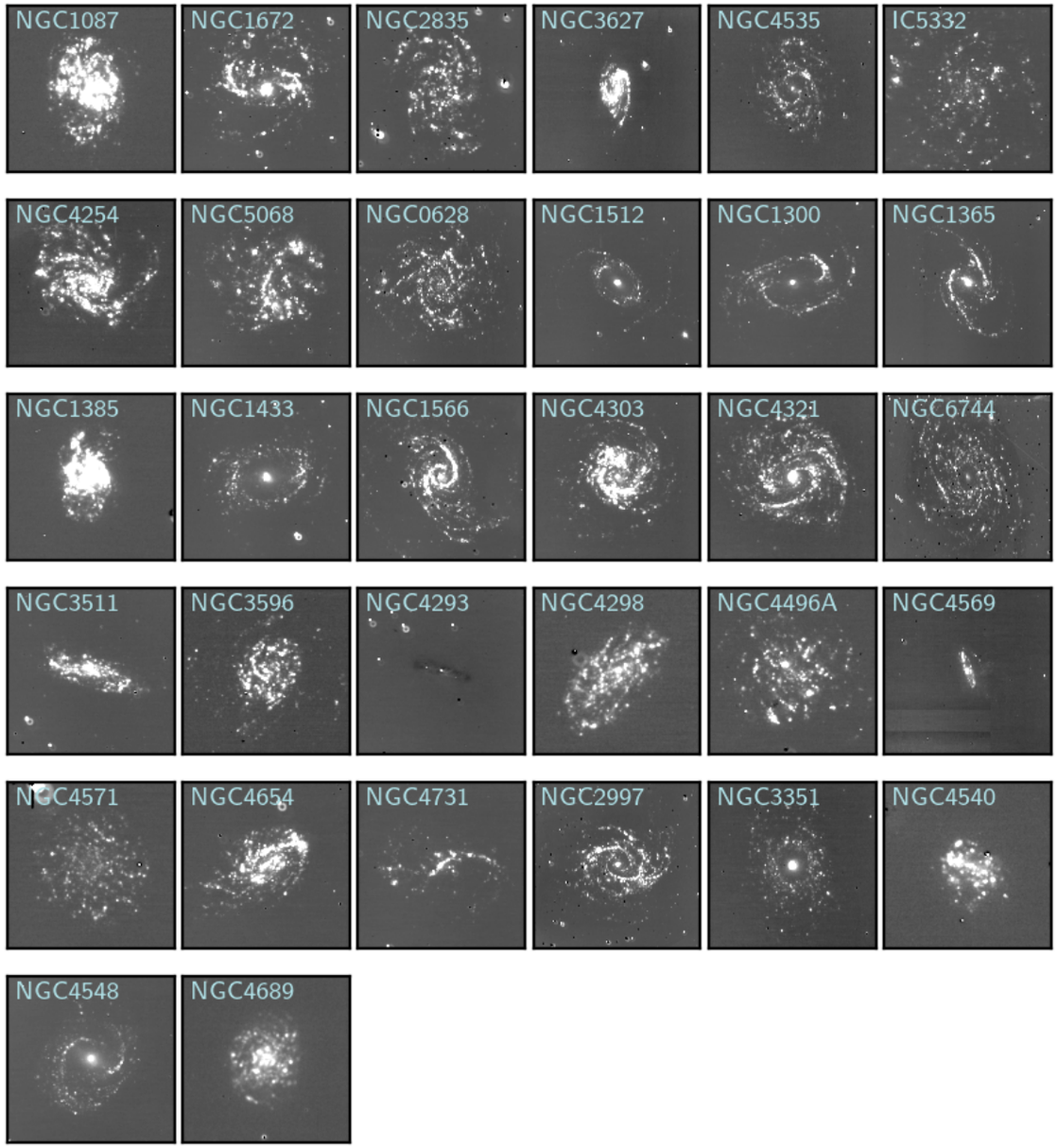
32 galaxies observed (~85% in service mode)

DirectCCD at 2.5-m duPont tel. Las Campanas

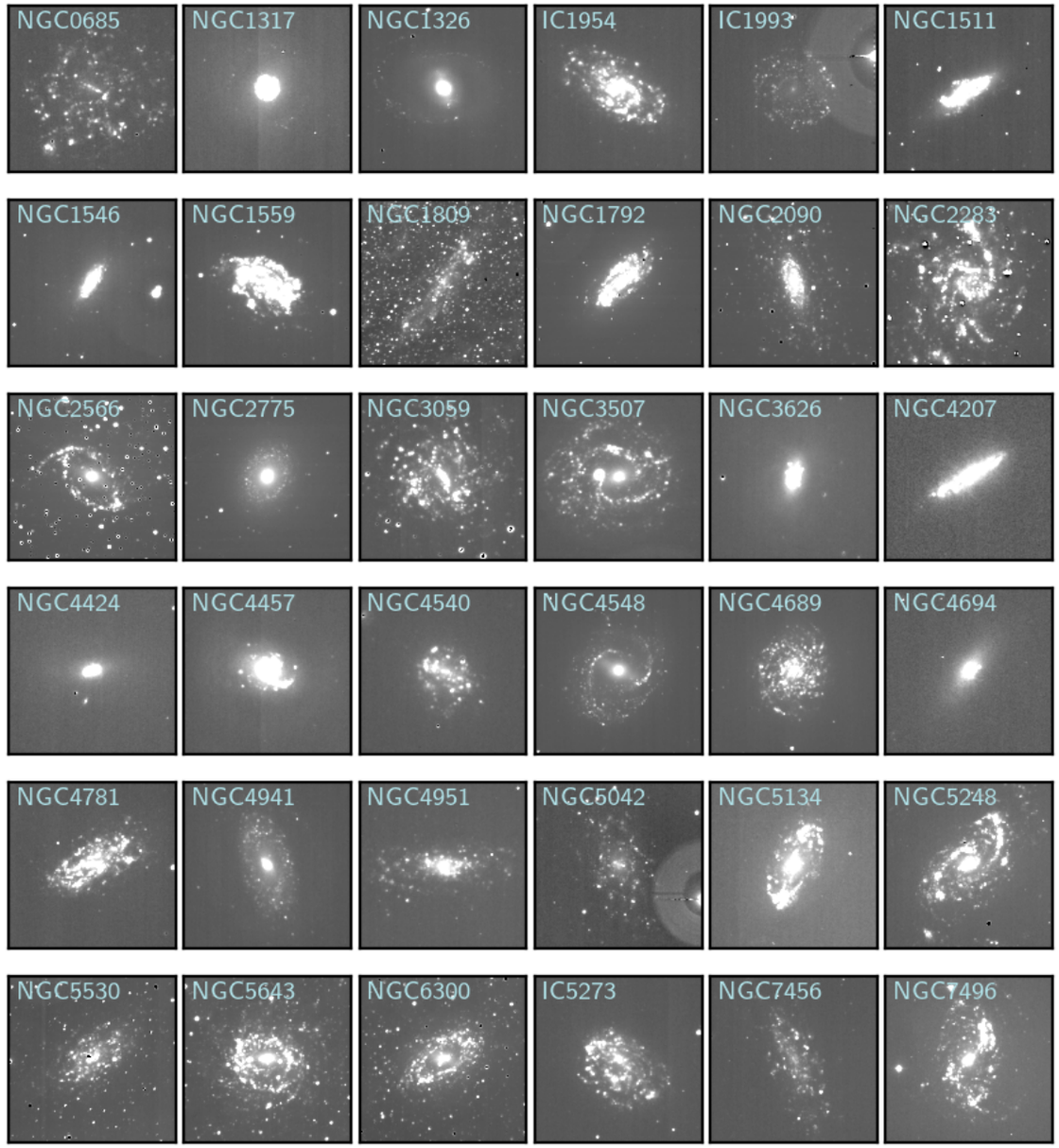
FOV 8’.85x8’.85 - 0.259 arcsec/pixel

36 galaxies observed

MPG2.2m/WFI galaxies



duPont/DirectCCD galaxies



Observing strategy

- Targets are observed in Hα narrow-band and R broad-band filters during the same night

- Standard dither pattern adopted

- A series of shorter exposures are taken, when necessary, to avoid saturation in the cores

Pipeline Overview

1. **CCD Detrending**
   CCDred_basic.py

2. **Astrometry**
   astromatic_pipeline.py

3. **Photometry**
   BB_phot_calib.py
   NB_phot_calib.py

4. **Background subtraction**
   PSF_compute_gauss_to_gauss.py

5. **Weighted combination**
   weight_combine_v23.py

6. **Continuum subtraction**
   Brent's code

Data preparation

1. A first manual step to create the *objects_txt*

This is needed for the rough astrometric solution

Try to use *make_objects_txt_WFI.py* or do it completely manually

2. Raw data in the right folder

Organize the raw data in your working folder by dividing them in observation dates (ex. 2022-10-23)

Running the codes

argparse **enables a simple help by writing the argument** *-h* **in the command line —> USE IT!**

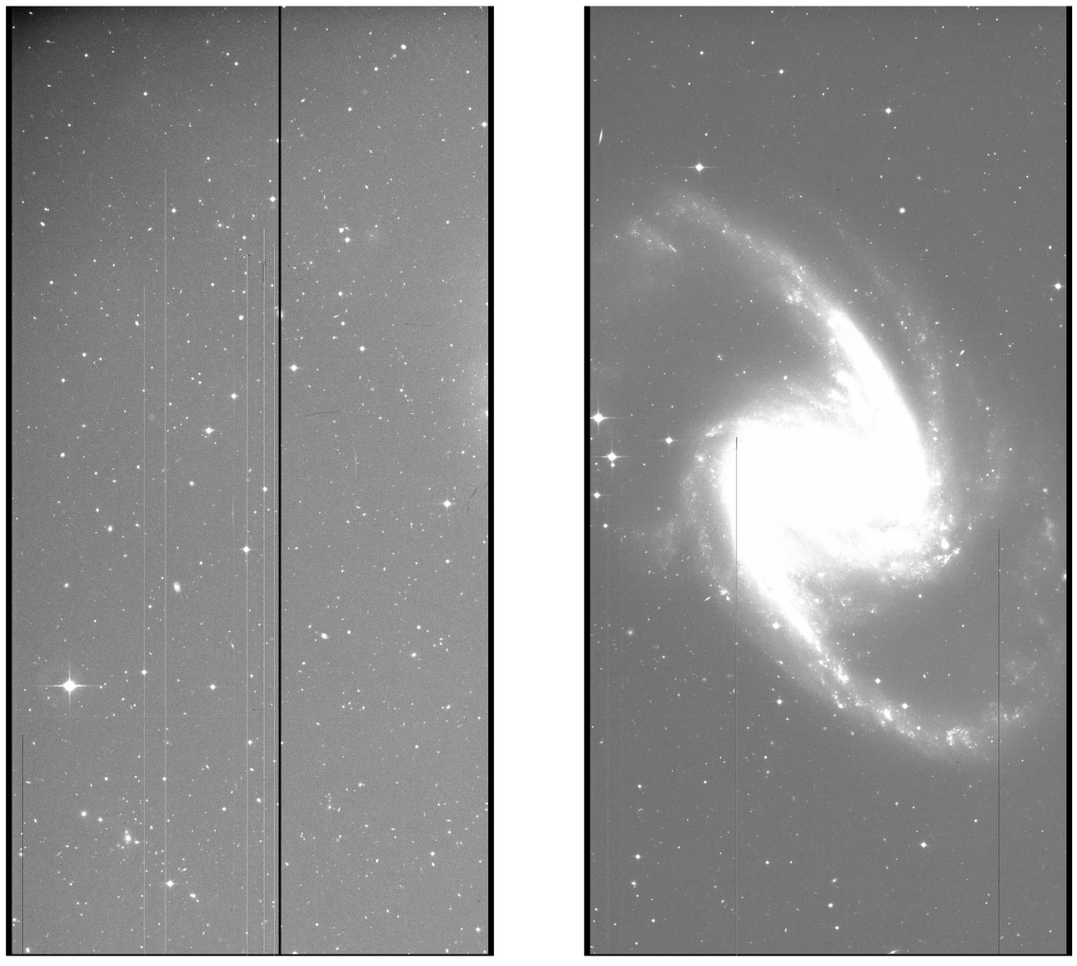
Folders are created in each step where the pipeline puts the output files. The same files are used for the subsequent step.

Check plots are also created for photometry and background subtraction.

CCD Detrending

Image detrend (a lot of pain)

Standard …

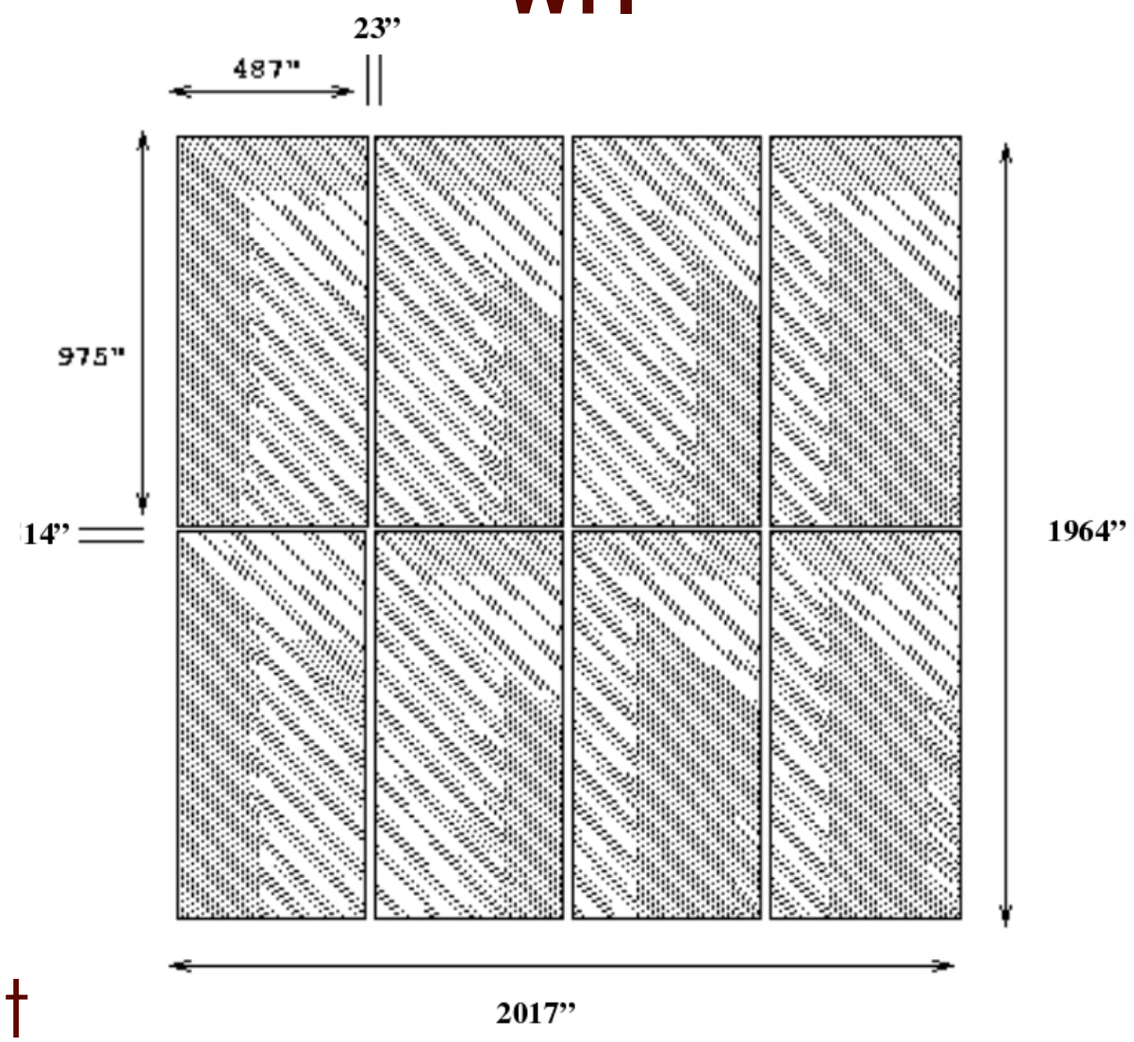


… and non-standard detrend recipes

Each CCD separately detrended

- Overscan subtraction + trimming
- Bias subtraction + flat-fielding
- Cosmic rays removal
- Gain correction + STD maps
- Bad pixel mask from master flat
- Normalization to electron/sec

WFI



Coverage: 0.56x0.54 deg^2 (0.238 "/pix)

Run CCD detrend

CCDred_basic.py pathtodata obsdate -o NGC0628 -jmebd

Input

Raw data

Output

masterbias<ccd>_corr.fits

masterflat_<r/h><ccd>_corr.fits

rwcs<galaxy>_<filt>_<ccd>_frame<num>.fits

Astrometry

WCS registration steps

- Rough astrometric solution (CCD detrend)
- **SExtractor** with faint and bright sources rejection (Astrometry)
- **SCAMP** registration of each CCD towards (Astrometry)
- **SWarp** to reconstruct the mosaic of CCDs for each science, mask and error frames (Astrometry)
- Offset between Gaia DR2 and the data applied (Post processing)

Run astrometry

WFIpipeline.py pathtodata obsdate -o NGC0628 -a -c

(*WFIpipeline.py* can call *CCDred_basic.py* to perform CCD detrend and astrometry together)

Input

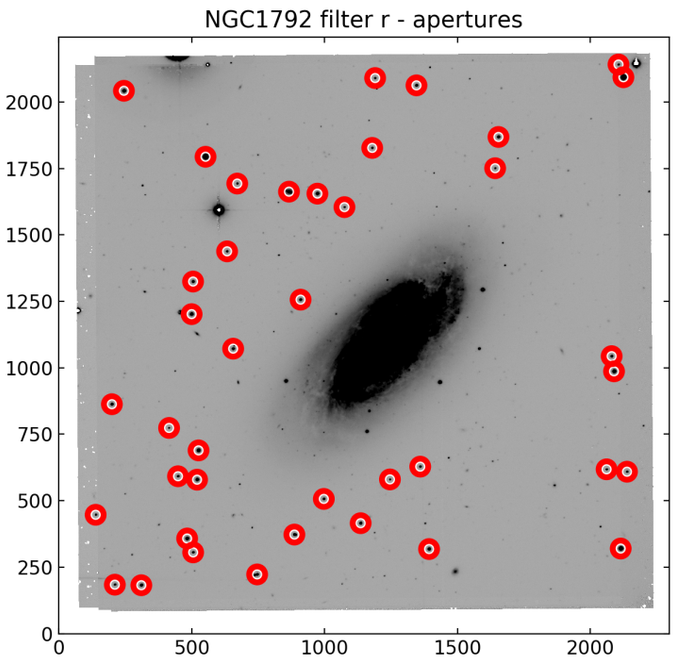
rwcs<galaxy>_<filt>_<ccd>_frame<num>.fits

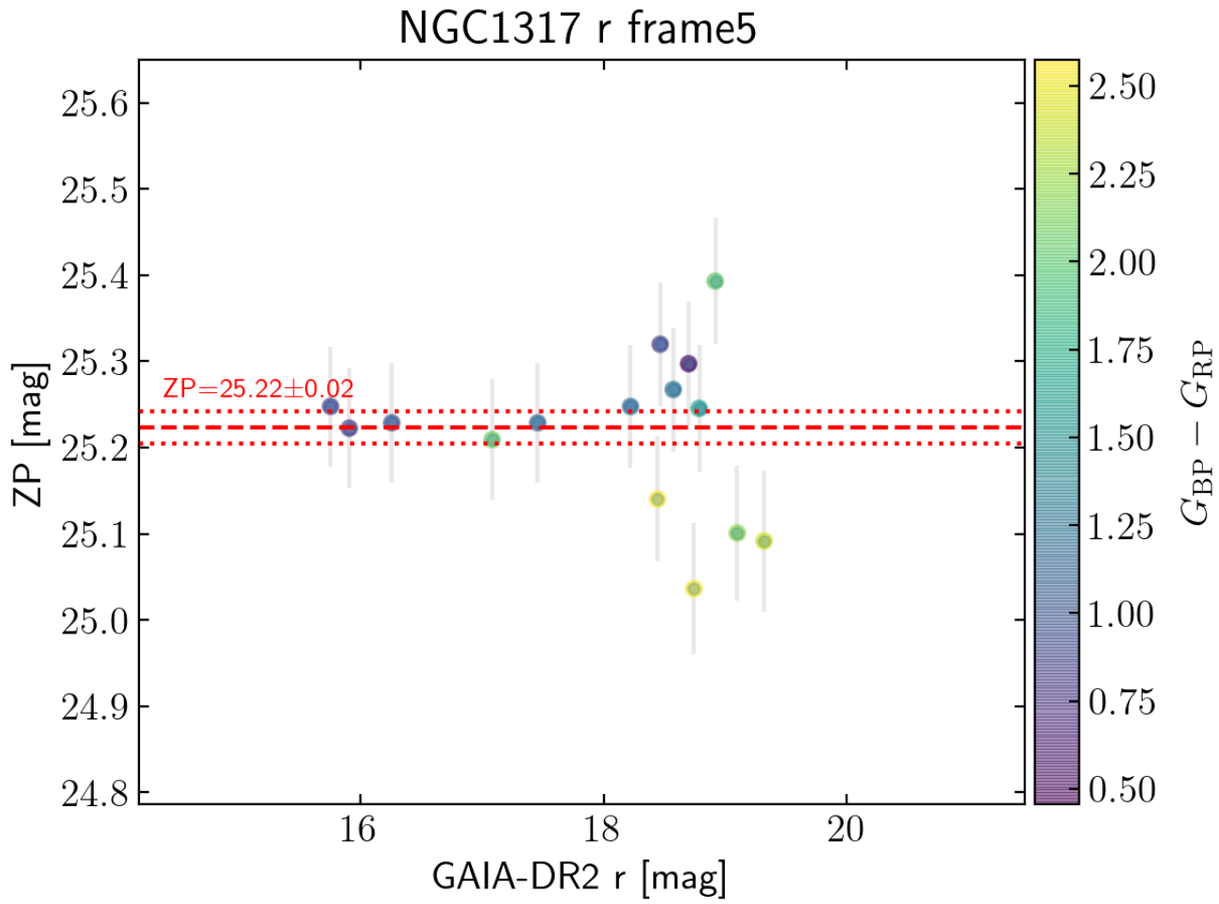
Output

<galaxy>_<filt>_stack<num>.fits
<galaxy>_<filt>_stack<num>_var.fits

Photometry

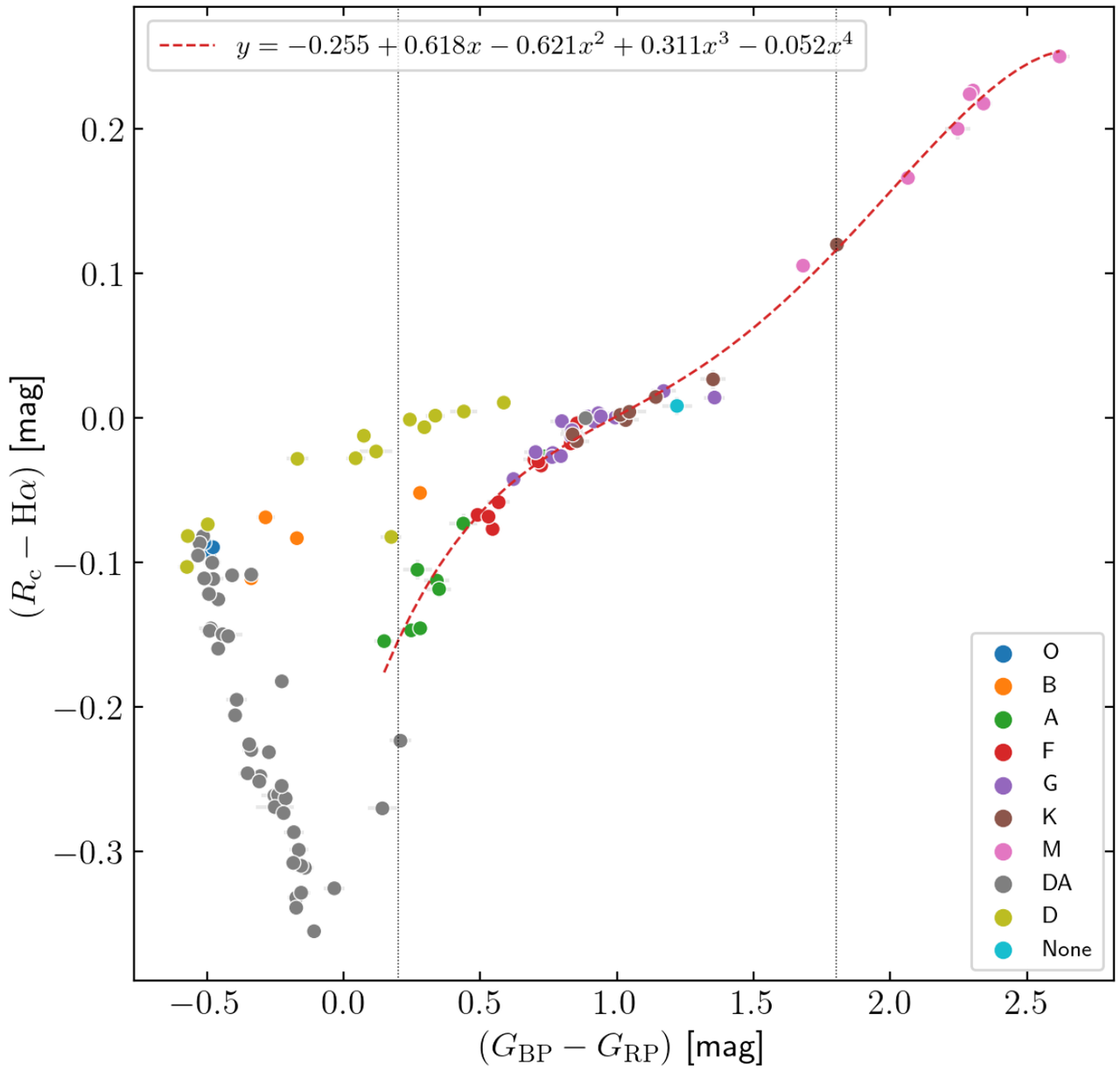
BB calibration





Stars are detected with a IRAFFind-like routine, then selected by excluding those too close to each other, to the boundaries and too faint (with a SNR cut), but also stars within galaxy ellipse.

NB calibration



Same detection, selection and aperture photometry as for the BB calibration, but a color-color relation is used to calibrate NB against the BB.

Run photometry

BB_phot_calib.py -w <NGC0628> -c <NGC0000>

NB_phot_calib.py -w <NGC0628> -c <NGC0000>

- Write the list of WFI object to reduce and set CCD (du Pont) objects to a non valid name
- Either use the argument of —wfi_path or edit the file *path_to_uncalib_WFI.txt*
- Leave default values for FWHM, SNR and elliptical radius for the galaxy, but they can be fine tuned

Input

<galaxy>_<filter>_stack<num>.fits

Output

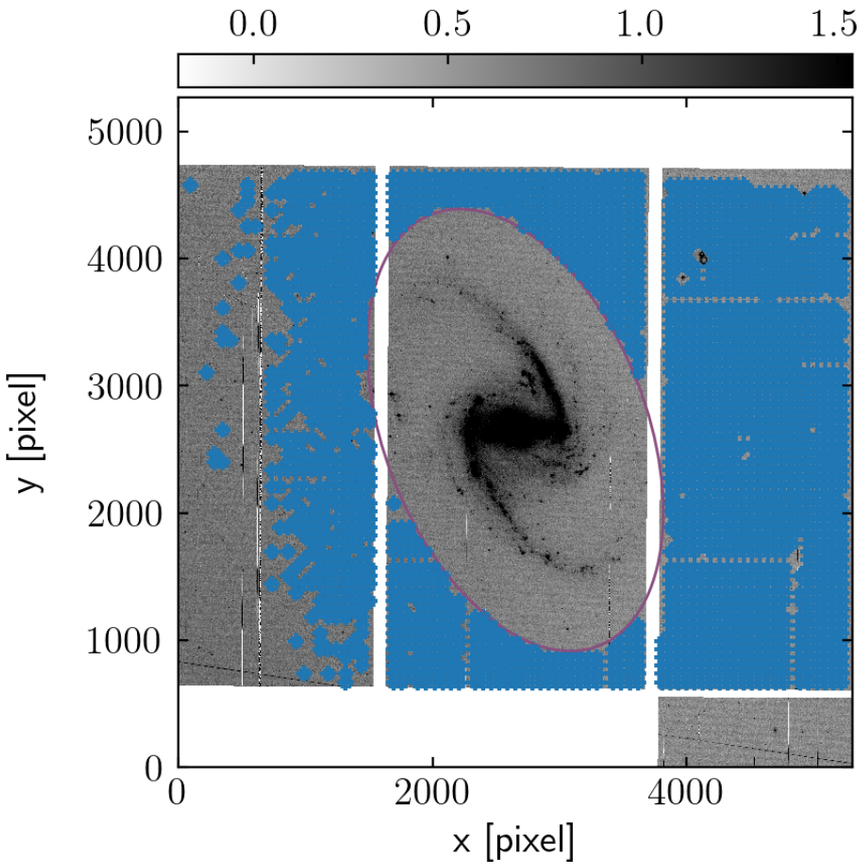
<galaxy>_<filter>_stack<num>_calib.fits

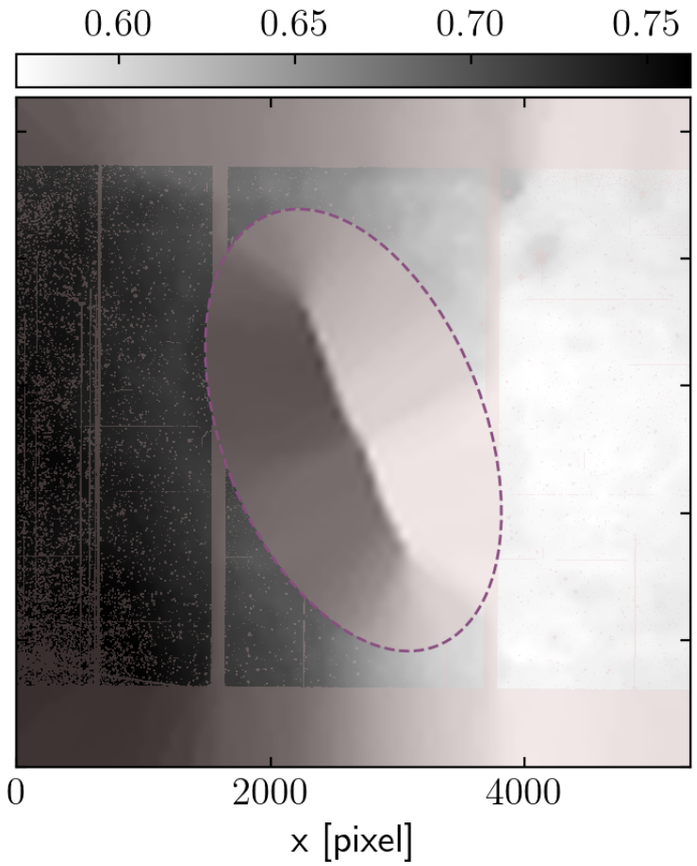
<galaxy>_<filter>_stack<num>_std_calib.fits

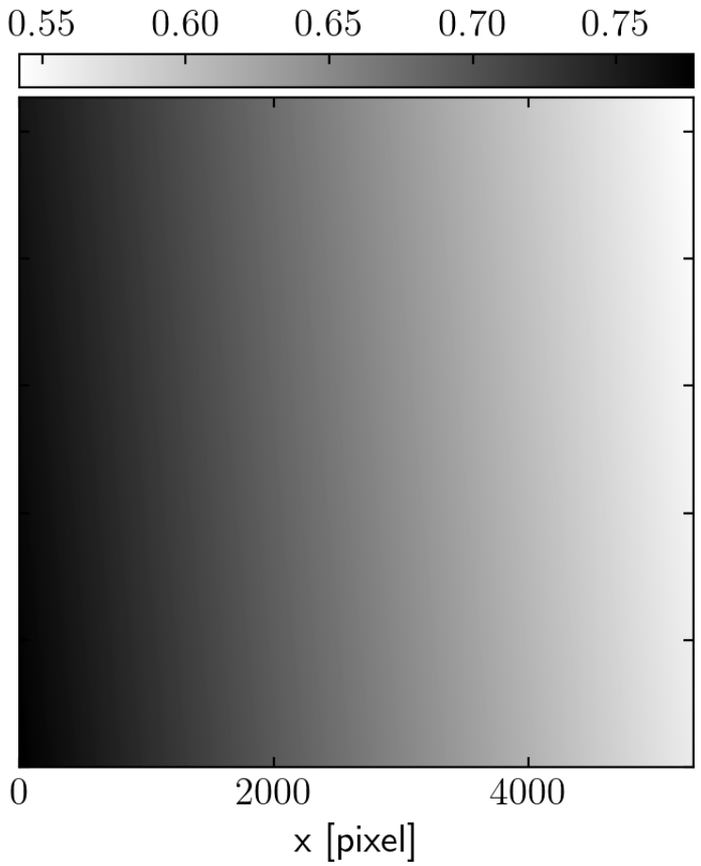
<galaxy>_<filt>_stack<num>_mask_calib.fits

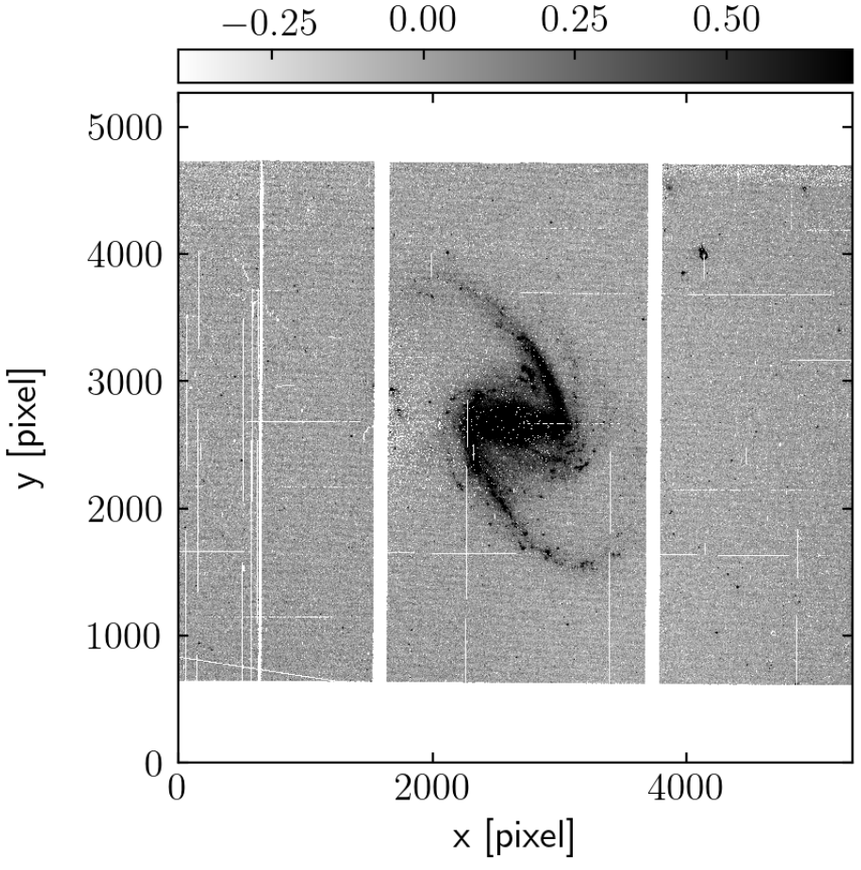
Forgot newmask and sp_mask!!
(edited after the tutorial at the end of this presentation)

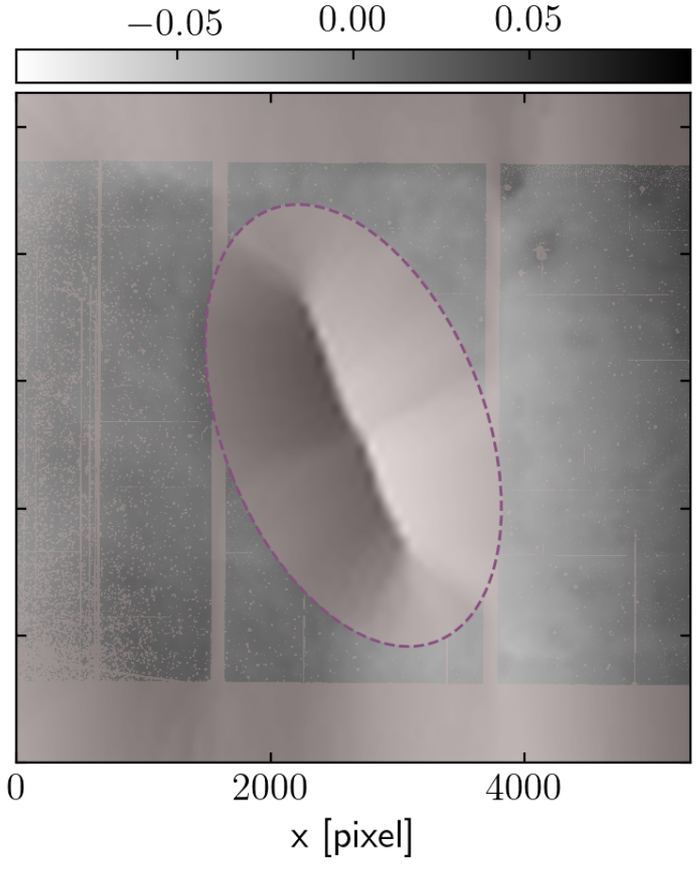
Background subtraction
&
weighted combination

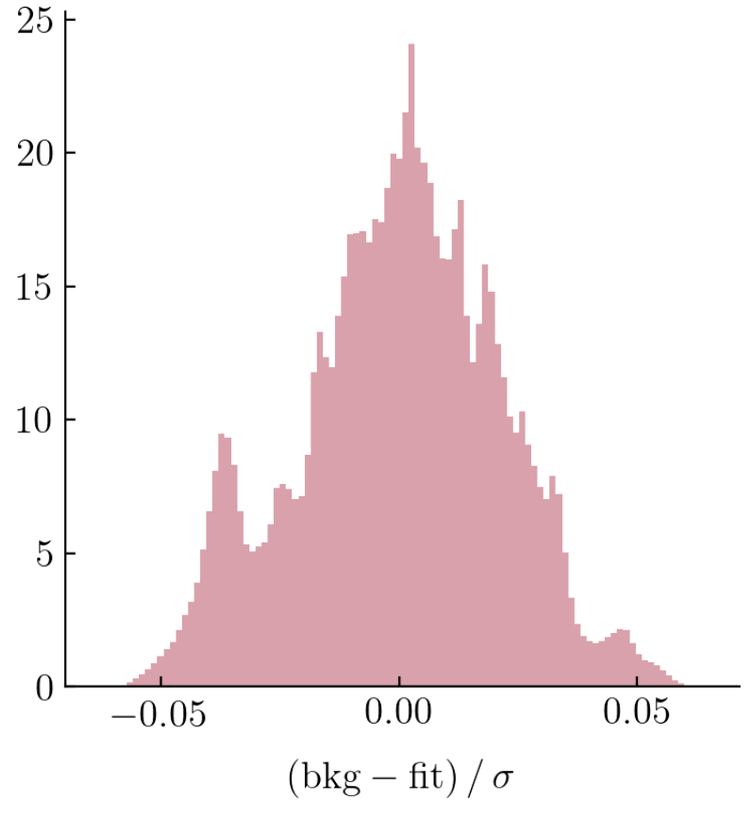

0.0
0.5
1.0
1.5
5000
4000
3000
2000
1000
0
y [pixel]
0
2000
4000
x [pixel]


0.60
0.65
0.70
0.75
0
2000
4000
x [pixel]


0.55
0.60
0.65
0.70
0.75
0
2000
4000
x [pixel]


−0.25
0.00
0.25
0.50
5000
4000
3000
2000
1000
0
y [pixel]
0
2000
4000
x [pixel]


−0.05
0.00
0.05
0
2000
4000
x [pixel]


25
20
15
10
5
0
−0.05
0.00
0.05
(bkg − fit) / σ

Run background subtraction

PSF_compute_gauss_to_gauss.py -w <NGC0628> -c <NGC0000> —bkg_subtraction

- Write the list of WFI object to reduce and set CCD (du Pont) objects to a non valid name
- Either use the argument of —wfi_path or edit the file *path_to_calib_WFI.txt*
- Leave default values for FWHM, SNR and elliptical radius for the galaxy, but they can be fine tuned
- Only the science frames are background-subtracted

Input

<galaxy>_<filter>_stack<num>_calib.fits

Output

<galaxy>_<filter>_stack<num>_calib_nosky.fits

Run weighted combination

weight_combine_v23.py —wfi_path -w <NGC0628> -c <NGC0000> —wfi_galaxies <NGC0628 NGC1365> —ccd_galaxies <NGC0000> —wfi_filters <filternames>

- Write the list of WFI object to reduce and set CCD (du Pont) objects to a non valid name
- You can decide to weight-combine the images for both filters (leaving default value for —wfi_filters)
- Either use the argument of —wfi_path or edit the file *path_to_uncalib_WFI.txt*

Input

<galaxy>_<filter>_stack<num>_calib_nosky.fits
<galaxy>_<filter>_stack<num>_std_calib.fits

Output

Read the first paragraph of README file at *PHANGS Google Drive*

Further data reduction steps (edited after the tutorial)

Final Remarks (edited after the tutorial)

Other two files hav e to be run independently and after the photometric calibration:

1. mask_reprojection_dcrzc.py

This was meant to update the bad pixel mask in a subsequent release without re-running the detrend step in the cluster. It reads *bad_columns.txt* and retroject the mask to a file with the string "_newmask" within its name.
Various short bad pixel columns are seen in longer exposures therefore we created a new mask that could be updated if necessary.

Final Remarks (edited after the tutorial)

Other two files hav e to be run independently and after the photometric calibration:

2. mask_saturated_pixels.py

This script updates the mask with the saturated pixels.
It was not included in a first release, therefore we added this script to update the mask with a new file with the string “_sp_mask” in its name.

Final Remarks (edited after the tutorial)

It would be good to check carefully the individual frames before the weighted combination step in order to detect satellite or asteroid tracks (the last are visible by blinking the frames in order) or visible bad columns and cosmic rays or even saturated stars spikes.

Masks with DS9 can be created to be read by the script.
The file name is expected to be
<galaxy>_<filter>_stack<num>.reg
and be put in a folder called *ds9_regions*

Final Remarks (edited after the tutorial)

A comprehensive list of auxiliary files is beyond the scope of this tutorial, therefore do not hesitate to ask if you need to know what a specific script does or what an auxiliary file is meant to be for..

Happy Data reduction!!